

Orbital Determination of Mars Crossing Asteroid 5066 Garradd

C. ALLEN, M. BEJARANO, D. RAMESH¹

¹ *The Summer Science Program, University of Colorado Boulder*

ABSTRACT

This paper discusses the process used to study the motion of the Mars-crossing Asteroid 5066 Garradd (1990 MA) and the process used to redetermine and confirm its orbit. The study included determining the asteroid's position on five observation nights spread throughout the months of June and July of 2025. Using the positions of 5066 Garradd during three of these observations, the position (with respect to the Sun) and velocity vectors were calculated via the Method of Gauss. With these vectors, the six orbital elements – the semi-major axis (a), eccentricity (e), inclination (i), longitude of the ascending node (Ω), argument of perihelion (ω), and time of perihelion passage (T) – were calculated through a series of mathematical processes as well as their corresponding uncertainties using the Monte Carlo method. These elements were found to be $a = 1.937 \pm 0.000005$ AU, $e = 0.1550 \pm 0.000002^\circ$, $i = 41.48^\circ \pm 0.00007^\circ$, $\Omega = 82.27^\circ \pm 0.00003^\circ$, $\omega = 184.7^\circ \pm 0.005$, $T = 2459893.09 \pm$, $M = 354.4^\circ \pm 0.0002^\circ$. The orbital elements were later used to determine the asteroid's orbit. Consistency checks were successful, as the use of such orbital elements produced an accurate ephemeris (R.A. and Decl.) for the remaining two observations.

1. INTRODUCTION

Orbital determination is the process of analytically and mathematically computing the orbit of celestial objects using a series of astronomical observations and well-defined classical physics relationships (R. P. Baucells (2022)). One of the most widely used classical methods of orbit determination – and the one used in this study – was developed by Gauss in 1801 and relies heavily on physical and geometrical relationships (R. P. Baucells (2022)).

Although the concepts and classical methods of orbital determination have been thoroughly established, the advent of new technology and discoveries in astrophysics has allowed these methods to increase in accuracy and efficiency. Particularly, the increased interest in Mars-crossing Asteroids (MCAs) and near-Earth asteroids (NEAs) has led to more wide scale surveying, including the 1991 Palomar Planet Crossing Asteroid Survey (E. F. Helin (2013)), which intricately identified risks that NEAs pose and increased our understanding of their population, orbits, and potential impact hazards.

The analysis of the impact of specific factors – including the time interval between successive observations, changes in the specific orbital parameters of the object,

and the number of observations – has also greatly increased the accuracy and effectiveness of using classical methods to determine celestial orbits from Earth (C. M. Roithmayr et al. (2005)). In addition, technological advancements in the use of algorithms and simulations for orbital determination have improved the efficiency of calculating celestial orbits, particularly those of asteroids.

Through the use of computational physics, Gauss's method can be extrapolated to a wide array of celestial objects with more complex orbital parameters. Vast databases of known celestial information, like the Minor Planet Center (P. Herget (1947)) and JPL Horizons, have been developed and used to extensively test open-source orbital determination software to improve accuracy and applicability to other celestial objects (R. Makadia et al. (2025)). An increased predictability in the trajectory of asteroids also allows for insight to be made into celestial bodies that have remained relatively untouched since the beginning of the solar system. The wealth of scientific information associated with asteroids can be accessed more easily with the incorporation of accurate simulations and models of how our solar system functions as a whole.

One of the ongoing challenges in the field of asteroid orbit determination is the Yarkovsky Effect, which is caused by a subtle force that affects the orbits of asteroids, especially MCAs, due to uneven heating and cool-

ing from solar rays (X. Liu et al. (2024)). This effect can cause the orbit of the asteroid to drift over time, producing either an increase or decrease in the semi-major axis, and complicates the process of determining an asteroid’s long-term orbit. Ongoing research in the field of asteroid orbit determination allows researchers to record changes in orbital parameters caused by celestial phenomena, contributing to both an increase in scientific understanding of asteroid orbits and the phenomena that can cause orbital changes.

Over time, a goal that has developed in the field of asteroid orbit determination research has been deflecting the path of asteroids in order to prevent future collisions with the surface of the earth. The 2022 NASA’s Double Asteroid Redirection Test (DART), for example, was a successful defense test that led to the intentional collision of a spacecraft into the asteroid Dimorphos in order to alter its orbital period (A. Buscarino et al. (2023)). This test demonstrated a viable method for deflecting asteroids that pose a risk to the earth by altering the asteroid’s speed and trajectory. DART also allowed for a greater understanding of subjects such as the impact velocity of NEAs, momentum transfer, and orbital deflection laws, all of which are factors that need to be analyzed for ongoing and future NEA deflection missions (K. Wang et al. (2023)). These deflection missions offer a way to potentially mitigate the effects of asteroid collisions with the earth.

The new observational data obtained of Asteroid 5066 Garradd contributes to a more precise determination of the asteroid’s orbit, improving tracking capability and an understanding of MCAs movement. The study serves to refine orbital predictions in the face of persistent uncertainties such as the Yarkovsky effect and limitations in the long-term predictability of asteroid trajectories. The study is particularly relevant for objects like 5066 Garradd which, although not considered hazardous, have a relatively large size of 3.8-8.6 km diameter (J. Mou & I. Webster (2021)) and classification as a Mars crosser, elevating our concern and its relevance for orbital path tracking. By integrating this study’s findings into the broader MCA catalog, we contribute to a global effort aimed at tracking and ultimately mitigating potential asteroid threats.

2. OBSERVATIONS

2.1. Target Selection

In astrophysical orbit determination, the selection of an apt target is paramount to the study as a whole. Asteroid 5066 Garradd was chosen for this study for a multitude of reasons. Garradd is classified as a Mars-crossing asteroid, meaning that its orbit brings the as-

teroid relatively close to earth in comparison to other astronomical distances. This results in the asteroid being a prime candidate for viewing from earth. More specifically, the asteroid chosen needed to be visible from the observation location of Sommers-Bausch Observatory at times optimal for viewing. This is aligned with Garradd’s projected R.A., Decl. values, and transit time of between 4:00 UTC and 5:00 UTC (22:00 and 23:00 MDT) as predicted by the data of JPL Horizons (Hor (1996)) and the Minor Planet Center (P. Herget (1947)).

Along with its ideal transit time, Garradd’s moderately-fast apparent velocity, with an average motion of $1.75''/\text{min}$ during the observed window, allowed for it to be easily detectable in both series and long exposure science images. The substantial change throughout an observation period as well as between different observation periods allows for a variety of valuable data to be calculated and eventually used in the calculation of orbital parameters.

The apparent magnitudes predicted by JPL Horizons ranged between 16.1-16.7, which lies within the optical range of the instrumentation available for the study, making it able to be detected during image processing (Hor (1996)). This, along with its movement patterns, ultimately ensures that – weather permitting – the asteroid is visible to the telescope and the camera collecting our science images. The combination of the above factors compile to make Asteroid 5066 Garradd a prime target for collecting clear and accurate data to be used in this study.

2.2. Data Acquisition

This study took place at the Sommers- Bausch Observatory, University of Colorado Boulder (MPC Code 463), with coordinates 40.0036° N , 105.2630° W (20-inch PlaneWave Telescopes, n.d.) The telescopes used were two 20-inch PlaneWave Telescopes, Artemis and Apollo, (CDK20, 20-inch, 0.51 meter) (20-inch PlaneWave Telescopes, n.d.). Affixed to the back of each telescope was a ZWO ASI6200MM Pro detector, which was used to capture the science images (20-inch PlaneWave Telescopes, n.d.).

The study involved a total of five observation nights, three of which are being used for orbital determination of Asteroid 5066 Garradd. Standard calibration techniques including dark and flat field corrections were applied.

Three series of light images were then taken at a 30 to 35 second exposure time for each image. In between each series, a 300-420 second long exposure was taken in order to assist in the identification. R.A. and Decl. values for the asteroid in each image were found using an astrometry and plate-solving process through the

Nova.astrometry.net web service. Observational parameters including date and time of observation, R.A. (α)

and Decl. (δ) of asteroid, and exposure time are presented in Table 1.

Table 1. Date/Time of Observation Midpoint in UTC and Julian Date, Asteroid R.A. and Dec, and Image Exposure Time.

Midpoint Date, Time (UTC)	Julian Date	α (HH:MM:SS.SS)	δ (sDD:MM:SS.SS)	Exp. Time (s)
2025-06-27 at 04:56:15.786	2460853.7057383	15:23:57.49	-07:25:59.79	30.0
2025-07-06 at 04:28:12.846	2460862.6862598	15:13:27.94	-13:46:19.99	32.0
2025-07-08 at 04:20:38.897	2460864.6810058	15:11:47.53	-15:05:25.65	32.0
2025-07-11 at 04:18:05.506	2460867.6792304	15:09:43.03	-17:00:15.25	35.0
2025-07-16 at 04:29:56.871	2460872.6874638	15:07:23.62	-20:01:09.85	35.0

Note: Asteroid R.A. is provided in hours : arcminutes : arcseconds format. Asteroid Decl. is provided in $\pm$ degrees : arcminutes : arcseconds.

2.3. Image Reduction

In order to provide accurate scientific data, science images collected during the observation stage were calibrated through software | using the aforementioned flat field and dark images | to adjust for discrepancies in the capturing of images by the CMOS. AstroImageJ was used to calibrate | with standard image reduction conventions | and plate-solve images through an Nova.Astrometry.net server in order to obtain accurate R.A. and Decl. values for the asteroid for each science image. Errors in the astrometry and plate-solving process were calculated when plate-solving occurred and were saved in a corrections file that was used during the Monte Carlo uncertainty calculation to provide uncertainty values for each orbital element.

3. METHODS

The Method of Gauss was used to determine the orbital parameters of Asteroid 5066. Developed in 1801, the Method of Gauss uses data from three observation periods to calculate the position vector and velocity vector for the asteroid relative to the second observation period. Given three sets of data – each including a R.A., Decl., time value for the observation of the asteroid, and the sun vector corresponding to that time – the Method of Gauss uses Kepler’s Laws and geometric relationships between the earth-sun vector ($\vec{R}$), earth-asteroid vector ($\vec{\rho}$), and sun-asteroid vector ($\vec{r}$) – visualized in Figure 1 – to determine a specific position and velocity vector at a singular point in time. Using linear algebra and the fact that all 3 position vectors lie in the same plane, the second position vector can be calculated using multiplicative transformations with scalar coefficients of a_1

and a_3 :

$$\vec{r}_2 = a_1\vec{r}_1 + a_3\vec{r}_3$$

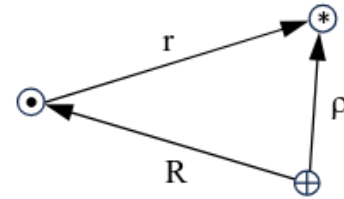
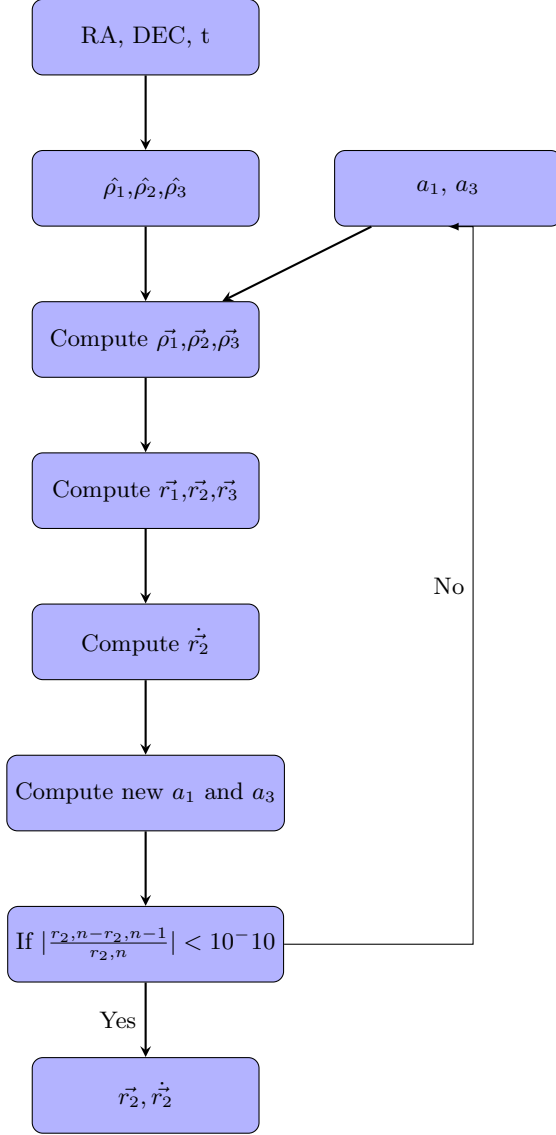


Figure 1. Visualization of the three vector quantities used in the Method of Gauss. The $\odot$ represents the Sun, the $\oplus$ represents the earth, and the $\ast$ represents the asteroid. Vector quantities $\vec{R}$, $\vec{\rho}$, and $\vec{r}$ are denoted accordingly.

Using the combination of the position vectors determined for the first and third observation periods and known R.A. and Decl. values, the general values for a_1 and a_3 and unit vectors for $\hat{\rho}$ are calculated respective to each observation period. These can be used to calculate the initial $\vec{\rho}$ vectors using specified linear algebra relationships. The ensuing calculations produce refined values for the position and velocity vector of the central epoch. With a new set of data produced with these calculations, an iterative process begins that recalculates the a_1 and a_3 coefficients, $\vec{\rho}$ vectors, $\vec{r}_2$ vector, and $\dot{\vec{r}}_2$ vector until minimal change is produced with the next iteration. The result is the final determination for $\vec{r}_2$ and $\dot{\vec{r}}_2$ that are utilized to calculate orbital parameters through a series of mathematical processes and properties of ellipses (M. Dubson (2024)). An outline of the process is pictured in Figure 2.

Figure 2. Flow Chart for the Method of Gauss Orbit Determination



Note: Simplified iteration steps of the Method of Gauss to compute the $\vec{r}_2$ and $\dot{\vec{r}}_2$ from R.A., Decl., and time (t)

In order to get accurate vectors and orbital parameters using the Method of Gauss, observation periods should be relatively spaced out – preferably with more than seven days in between observations for accuracy – and should have a relatively similar period of time between them for ease of calculations (M. Dubson (2024)). For this study, observations from 2025-06-27 (Night 1), 2025-07-06 (Night 2), and 2025-07-16 (Night 5) were chosen due to the time difference of greater than seven-days between all three observations as well as the team

consensus of clear observation conditions. For these observation dates, the second image from the second set of science images and corresponding data was used in all of the calculations during the process. Utilizing the data collected during these observation periods – as displayed in Table 1 – the position and velocity vectors of the central epoch of observation, and thus the orbital parameters, were calculated through the use of an iterative Python algorithm developed by the team.

Monte Carlo simulations were developed to account for and report the variations for the determination elements based on R.A. and Decl. variations. These simulations were also developed in Python, and were used to generate different orbital elements based on the R.A. and Decl. variations. Using the correspondence files between observed and cataloged R.A. and Decl. for each observation image during the plate-solving process from the Nova.astrometry.net web service, the standard deviation was calculated for the R.A. and Decl. values of the asteroid through a root mean square measurement. These standard deviations were then used to recalculate R.A. and Decl. values by adding a randomly generated value in the range from 0 to the standard deviation values of the R.A. and Decl., respectively. These new values were utilized to find a new position and velocity vector for the asteroid on the respective date, and then to create a histogram for each orbital element, containing 100000 iterations of recalculations. See Figure 3 on Page 6.

3.1. Consistency Check

With the orbital elements calculated, a Python ephemeris program developed by the team was used to calculate the projected R.A. and Decl. of Asteroid 5066 at other times. For these consistency checks, the team chose from the dates and times of other observation nights with the R.A. and Decl. of the asteroid determined. The calculated values of R.A. and Decl. were then compared to the observed and measured values, which shows an accuracy of within one arcsecond for both values – produced in Table 2. These deviation calculations attest to the accuracy of the calculated orbital parameters.

4. RESULTS

The results obtained from this study are the six orbital elements needed for the orbital determination of the asteroid, which include the semi-major axis (a), eccentricity (e), inclination angle (i), longitude of ascending node (Ω), argument of perihelion (ω), and the mean anomaly (M). The Monte Carlo uncertainties for orbital

values were also computed and recorded. These values are presented in Table 3, below.

Table 2. Self-Consistency Check

Date and Time	α Measured	δ Measured	α Calculated	δ Calculated
UTC	HH:MM:SS.SS	sDD:MM:SS.SS	HH:MM:SS.SS	sDD:MM:SS.SS
2025-07-08 at 04:20:38.897	15:11:47.53	-15:05:25.65	15:11:47.69	-15:05:25.91
2025-07-11 at 04:18:05.506	15:09:43.03	-17:00:15.25	15:09:43.18	-17:00:15.84

Note: R.A. is provided in hours : minutes : seconds and Decl. is provided in $\pm$ degrees : minutes : seconds.

Table 3. Orbital Elements, Mean Anomaly, and Monte Carlo Uncertainties using Observation Nights 1, 2, and 5.

Results	a (AU)	e ($^{\circ}$)	i ($^{\circ}$)	Ω ($^{\circ}$)	ω ($^{\circ}$)	T (JD)	M ($^{\circ}$)
Results 1	1.936391	0.154612	41.47559	82.26930	184.6962	2459893.206	355.1143
Results 2	1.940254	0.155992	41.49982	82.25705	184.7311	2459904.939	354.3751
Results 3	1.937379	0.154963	41.47725	82.26580	184.7036	2459893.089	354.3836
Uncertainties	± 0.000005	± 0.000002	± 0.00007	± 0.00003	± 0.0003	± 0.005	± 0.0002

Note: Although results from all team members are depicted, Monte Carlo uncertainties are calculated using Results 3.

Figure 3 on page 6 presents a set of histograms showing the distribution of orbital elements that result from the Monte Carlo simulations, one diagram for each of the orbital elements.

5. DISCUSSION

As a whole, this research study and the data gathered was extremely useful and successful. The orbital elements are all within 1% of the corresponding values in NASA’s JPL Horizon’s System, showing a high accuracy and consensus when compared to other orbit projections. The R.A. and Decl. coordinates obtained for other observation dates are within 1 arcsecond of recorded values, which highlights the accuracy between the orbit determined and strictly observational results gathered by the team.

Sources of error during the study might include calibration or plate-solving discrepancies that would cause determined R.A. and Decl. to be less accurate, resulting in differing position and velocity vectors and ultimately differing orbital elements. Repetition and comparison of the calibration and plate-solving process could improve

the accuracy of the results. Five comparison stars were used per night, and linear regression was performed for detection of variability and outliers, but the use of different comparison stars when repeating the calibration and plate-solving process could also improve accuracy of results. Observation conditions could also lead to variations in the accuracy of images collected during observation nights, leading to differences in determined R.A. and Decl. altering further calculations.

6. REFLECTION

6.1. Individual Reflections

6.1.1. C. Allen:

This research study was extremely enriching both academically and personally. I learned so much over the course of this study that I would not have had the opportunity to otherwise. I thoroughly enjoyed working on this study using real research gathered by our team and the amazing instruments available to us here at the Sommers-Bausch Observatory. Within this short five-week time period, we were able to successfully observe

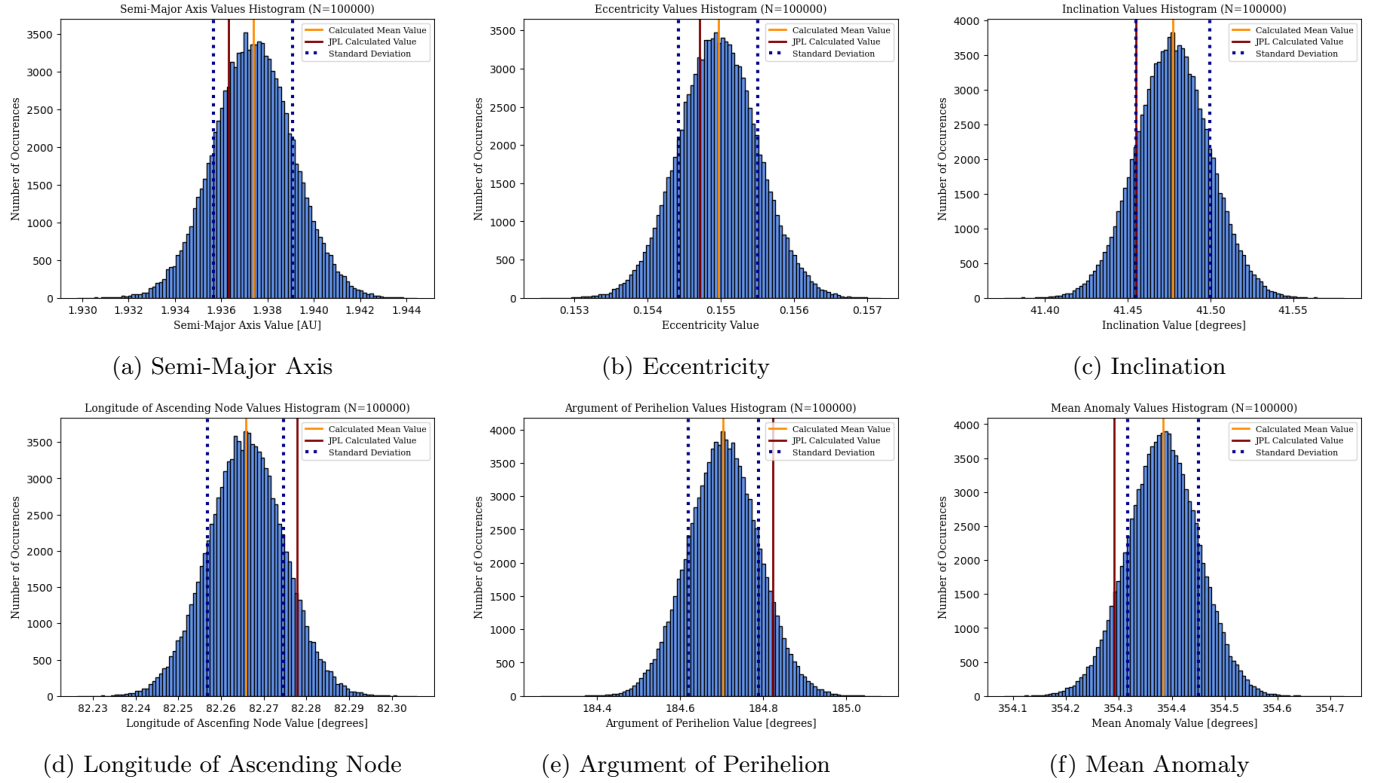


Figure 3. The normal distributions of the six orbital elements after accounting for the Monte Carlo uncertainties. Each graph displays 100 bins and 100000 iterations of recalculating the respective element.

and determine the orbit of our asteroid, quite the accomplishment for not knowing any undergraduate astronomy or physics prior to the program. I have become comfortable with what future research experiences may entail and now feel more prepared than ever to take on astrophysical research in the future.

6.1.2. *M. Bejarano:*

My experience doing this project was amazing. It was one of the most time-consuming and intensive research studies I have done, in great part because of the short time window in which we completed it. However, I greatly enjoyed learning Python programming for the study, and becoming very comfortable with it at the end of the research. I also enjoyed getting to work with real data, and the in-person work with the telescopes every night of observation. This was truly a dream for me, and a great opportunity in my future career path.

6.1.3. *D. Ramesh:*

This academic experience with the Summer Science Program was a very valuable experience and helped me grow as a researcher and a student. It was extremely intensive and informative, and I absolutely enjoyed working with a diverse set of teammates and experts, and gaining hands-on experience with using professional

instruments, such as the telescopes at the Sommers-Bausch Observatory. Overall, I felt that broadening my expertise in various domains of astrophysics helped me become more confident in my future as a researcher.

6.2. *Team Reflection*

As a team, we divided work efficiently. We individually completed Python code for the determination of orbital elements and compared the results for accuracy, and then we split the remaining work: C. Allen worked on developing the Monte Carlo simulations, while M. Bejarano and D. Ramesh worked on the written report and findings presentation. In the end, each team member had a very distinguished impact on the study as a whole. Our team worked very well together as we built co-working and teamwork skills and the research developed. We had very good coordination during observation nights, especially those toward the end of the study, and while working together on code and report writing. We wish we had established those working dynamics since the beginning of the research study, but we are proud of both the final result and the team dynamics we developed.

7. CONCLUSION

This study of 5066 Garradd (1990 MA)’s orbit was successful in redetermining and confirming its orbit. The values obtained differ slightly from the ones given in JPL, and consistency checks were successful throughout the study. This highlights the accuracy of the study and its implications in improving orbital determination of asteroids in the future. Future work includes the development of a three-dimensional model of the asteroid’s orbit, and possibly graphical representation of its comparison with Earth’s and Mars’ orbit. It is also planned to determine the absolute H (angular momentum) magnitude of the asteroid, with the purpose of obtaining a better understanding of its physical properties. The determination of the orbital elements of 5066 Garradd (1990 MA)’s contributes to a deeper understanding of the motion of MCAs, and might also help in the under-

standing of how they interact with other objects in the solar system.

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